

UVA K-12 ADVISORY COUNCIL · OCTOBER 2, 2026

Innovation with Guardrails

What AI can do for schools now, and how to decide where to use it

Brendan Bartanen · UVA School of Education and Human Development

Agenda

- 1 Where things stand
- 2 Two examples, both built this week
- 3 Five principles the examples show
- 4 How to decide where to use it

My guiding principles

- Maintain human judgment and expertise.
- Children learn from caring adults, not robots.
- Innovation and harm are both possible, so vigilance is essential.

AI is a general-purpose technology, like electricity

- Nobody asks whether a school is for or against electricity.
- We ask where it should go and what the building code requires.
- The same two questions work for AI: where to use it, and under what rules.

Factories took decades to benefit from electricity

- Edison's first power stations opened in 1882.
- At first, factories put motors on their existing shafts and belts. Productivity barely moved.
- The gains came in the 1920s, when factories were rebuilt with a motor on each machine.

Share of U.S. factory power capacity from electric motors

1899: under

5%

Early 1920s: just over

50%

David, P. A. (1990). The dynamo and the computer. *American Economic Review*, 80(2), 355-361.

Teachers and students already use AI

60%

of public school teachers
used an AI tool for work
in 2024-25

Gallup and Walton Family Foundation, 2,232
teachers, March-April 2025

54%

of teens ages 13 to 17
have used a chatbot for
help with schoolwork

Pew Research Center, 1,458 teens,
September-October 2025

18%

of teachers have formal
guidance from
administrators on using
AI

Gallup and Walton Family Foundation, 2,069
teachers, February-March 2026

Gallup (June 24, 2025; May 26, 2026). Pew Research Center, How Teens Use and View AI (February 24, 2026).

AI assistants can now complete long tasks on their own

2023

A chatbot answers a question or drafts a paragraph. A person does everything else.

2026

A coding tool such as Claude Code can work for hours on one request. It reads files, writes and runs code, checks its results, and reports back.

The longest software task an AI can finish has doubled about every seven months since 2019. Length means the time the task takes a skilled person.

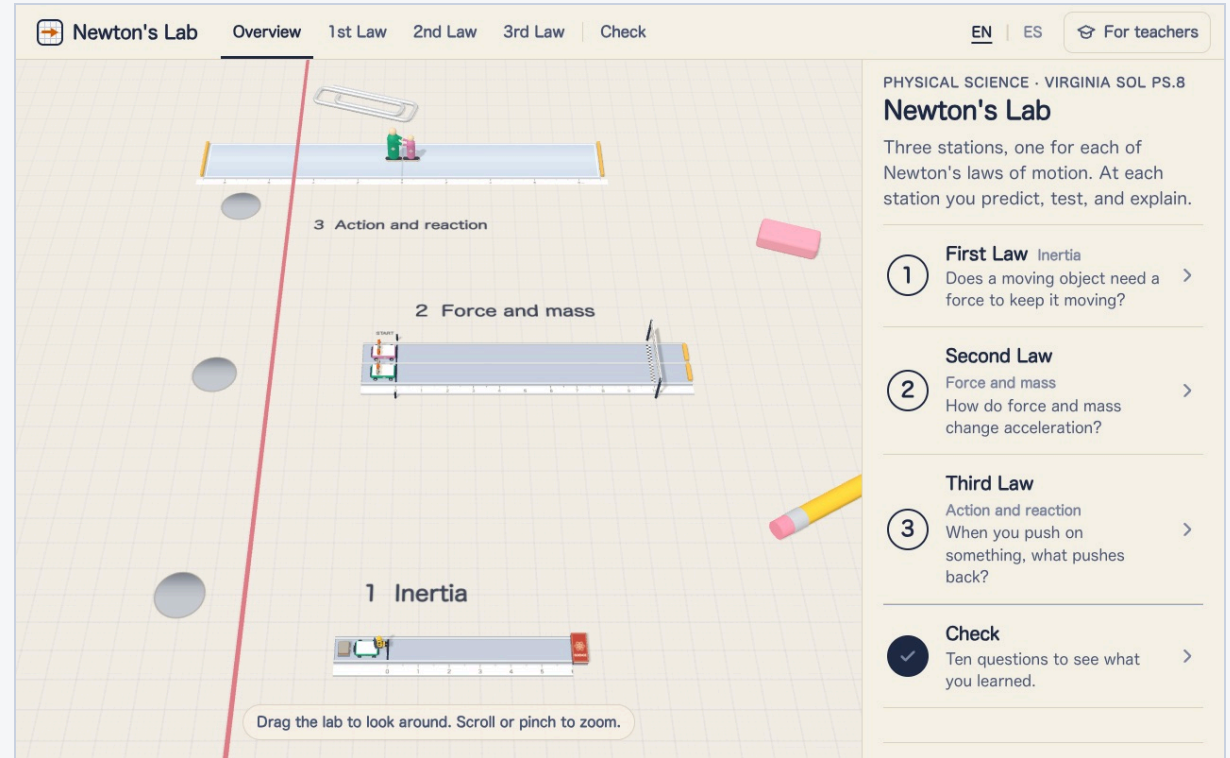
METR, Measuring AI Ability to Complete Long Software Tasks (2025) and Time Horizon 1.1 (January 2026).

EXAMPLE 1

A lab for 8th-grade science

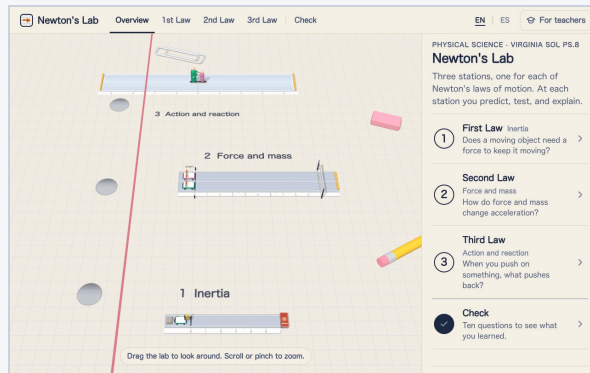
- I taught 8th-grade science from 2013 to 2015.
- On Wednesday night I asked Claude Code to turn my old lessons on Newton's laws into a 3D lab for students, aligned to Virginia's standards.

Open the lab



The lab was built overnight from a one-paragraph request

Wednesday night. I typed one paragraph. One AI assistant directed six others for about four hours. Result: the lab on a notebook page.



Wednesday night

Thursday morning. I asked for better graphics and suggested Scott Stadium. About three hours. Result: the stadium.



Thursday morning

Thursday afternoon. "Can we make the tracks longer?" About an hour and a half. Result: both cart tracks doubled, with cameras and tests updated.

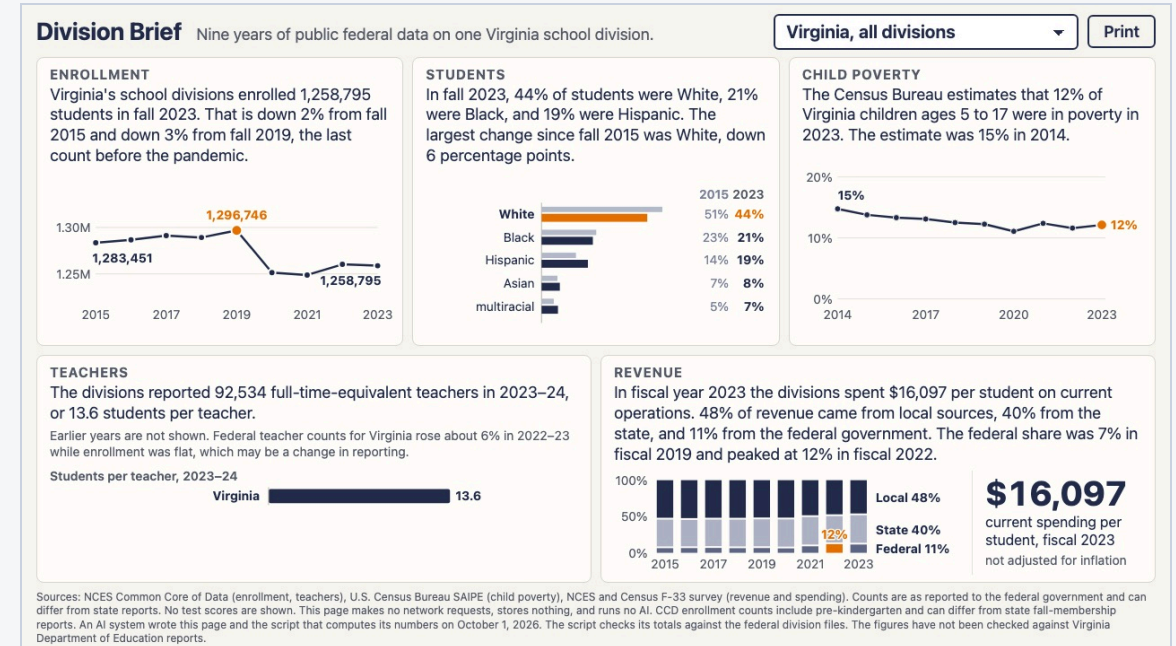
The lab is about **15,000** lines of code, and I wrote none of it.

EXAMPLE 2

A data dashboard

- Public federal data on any Virginia division.
- Built Thursday afternoon from data files I already had.

[Open the Division Brief](#)



Five principles from the two examples

- 1 Building software no longer requires writing code.
- 2 The scarce skill is judgment.
- 3 Teachers have the judgment and the context.
- 4 AI that builds is different from AI that talks to students.
- 5 AI is best at tasks whose answers are easy to check.

PRINCIPLE 1

Building software no longer requires writing code

- Until recently, new software needed engineers.
- I wrote none of the code and spent about 30 minutes.
- Building it took a coding tool and a frontier model.

Teachers will build the best educational uses of AI.

PRINCIPLE 2

The scarce skill is judgment

- Judgment is knowing what is worth building, what to do next, and whether the result is correct and useful.
- I decided what the lab should teach and what to leave out.
- The AI flagged about fifteen mistakes or shaky explanations in the old unit. A physics teacher still has to check each one.

PRINCIPLE 3

Teachers have the judgment and the context

- Teachers use judgment all day about their students.
- The AI should start from their lesson files.
- Custom versions are cheap: Scott Stadium took one request.

PRINCIPLE 4

AI that builds is different from AI that talks to students

AI builds, an adult reviews

- Students see a finished product.
- It can be tested and corrected.
- No AI runs when students use it.

AI talks to students

- Every answer is new.
- No adult sees it first.
- It needs limits, monitoring, and a plan for mistakes.

Ask of any proposal: does AI run while students use it?

PRINCIPLE 5

AI is best at tasks whose answers are easy to check

Checked by computer

- 69 automated physics tests
- Automated walk-through in English and Spanish at three screen sizes

Still needs a person

- Review by a physics teacher
- Review by a Spanish-speaking teacher
- A test on a school Chromebook
- A trial with students

Before trusting AI with a task, ask how you would know if the result were wrong.

Wei, J. (2025). Asymmetry of verification and verifier's rule.

How do we decide where to use AI?

Three questions to ask

- 1 Is doing the task the point?
- 2 Can a qualified adult check the result before anyone relies on it?
- 3 How costly is a mistake, and can it be undone?

Three questions before handing a task to AI

Question 1: Is doing the task the point?

Yes: keep it

- My lesson plan.
- Planning is when I work out where students will get stuck. When AI writes the plan, I teach a worse class.

No: it can be handed off

- The code for a simulation.
- I need a correct lab. Writing the code myself would add nothing to my teaching.

The same question applies to students and new teachers. When doing the task is how they learn, AI should not do it for them.

Two more questions for a task you hand off

2 Can a qualified adult check the result before anyone relies on it?

A mistake is cheap to fix

Yes, it can be checked first

Go.

The person using it checks.

No, it cannot be checked first

Pilot.

Set limits and review a sample afterward.

3 How costly is a mistake, and can it be undone?

A mistake is costly or hard to undo

Go, with sign-off.

One named person approves each output.

A person decides.

AI only gathers information.

Where would you put each of these?

		Yes, it can be checked first	No, it cannot be checked first
A A teacher has AI build a simulation for next week's science lab.	CLEAR	Go. The person using it checks.	Pilot. Set limits and review a sample afterward.
B AI drafts IEP goals from a student's records.	A mistake is cheap to fix		
C An AI tutor gives students hints during math practice.	A mistake is costly or hard to undo	Go, with sign-off. One named person approves each output.	A person decides. AI only gathers information.
D An AI detector flags a student's essay as AI-written.			

One reasonable sort

		Yes, it can be checked first	No, it cannot be checked first
B, C, and D all send student data to a vendor. Ask where it goes.	A mistake is cheap to fix	Go. The person using it checks. A lab simulation	Pilot. Set limits and review a sample afterward. C AI tutor
	A mistake is costly or hard to undo	Go, with sign-off. One named person approves each output. B IEP goals	A person decides. AI only gathers information. D AI detector

Divisions need staff with strong AI tools more than they need AI products

An AI product from a vendor

- It is built for many divisions at once.
- You cannot see or change it.

Your staff with a coding tool

- They build for your curriculum and students.
- They can test and change it.

You still pay for access to an AI model, under a data agreement.

Three things to try this month

- 1 Find the teachers already building with AI, and go see what they have made.
- 2 Give a few staff a frontier coding tool, under a data agreement, and time to use it.
- 3 Sort the AI requests already on your desk with the three questions.

Questions?

Please feel free to reach out to me at
bartanen@virginia.edu.

Try the two examples:
brendanbartanen-svg.github.io/k12-ai-examples



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